ALCHEMIX / ZETA-26

Dashboard Views Interpretation Report

A concise guide to every chart, metric, fact, and decision assumption across the five historical-analysis views.

REPORT SCOPE

HISTORICAL COVERAGE

Total observations **18,000**

Interplanetary links **12**

Historical ticks **500**

Traffic records **6,000**

Telemetry records **6,000**

Incident records **6,000**

Prepared from the supplied CSV histories and universe configuration | 06 July 2026

How to read the five views

The dashboard is descriptive evidence for the Analytical Co-Pilot. It helps explain historical behavior before trained models are used for live routing.

AVERAGE LOAD	MEDIAN LATENCY	TELEMETRY MAPE	JAM RATE	SATURATION
29.9%	**153k ms**	**8.9%**	**8.2%**	**3**
P95 61.3%	P95 465k ms	Median 3.0%	493 disrupted windows	257 latency values missing

What the dashboard can support

- Compare links under the same historical window.
- Identify non-linear congestion behavior.
- Prioritize suspected telemetry deception.
- Measure whether high-share routes are targeted more often.
- Explain a transparent risk index during a decision audit.

What the dashboard cannot prove

- Correlation alone does not prove Chimera caused a delay or jam.
- Historical risk does not guarantee the next live outcome.
- The composite score is not a trained prediction.
- Filtered views may have smaller samples and more unstable estimates.
- Null values must never be interpreted as zero latency.

SOURCE-TO-VIEW MAPPING

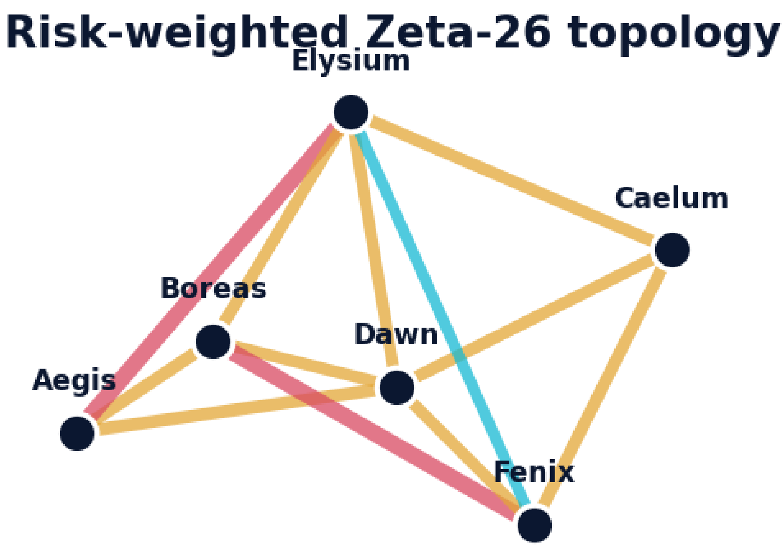
Traffic history	Telemetry history	Incident history
Congestion and latency	Trust and spoofing	Targeting and jamming

VIEW 01

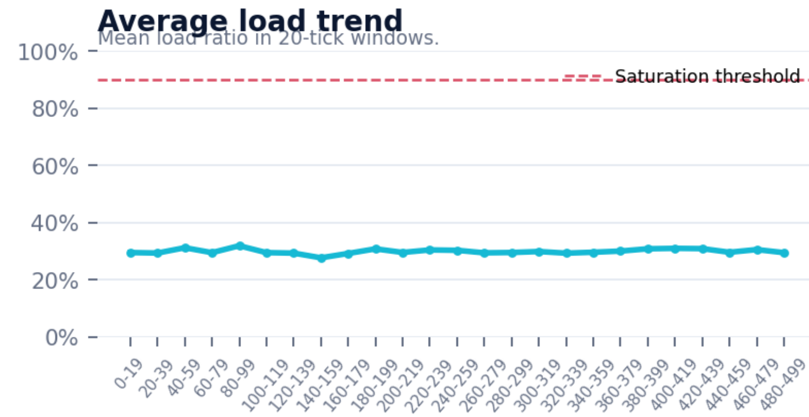
Command overview

A high-level operational picture: coverage, network risk, automatically generated facts, and broad time trends.

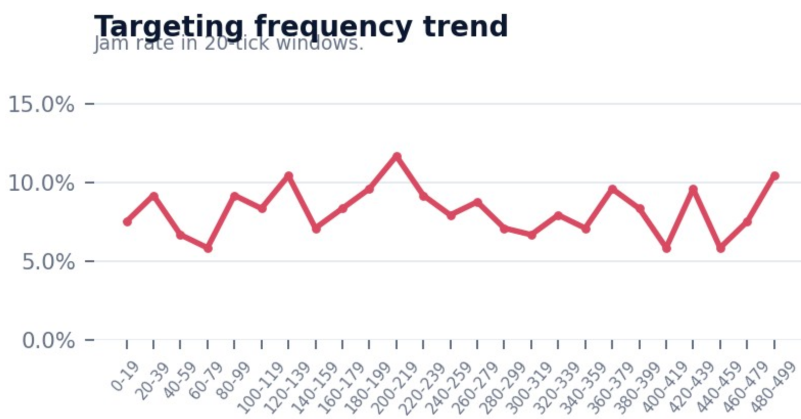
VISIBLE OBSERVATIONS	AVERAGE NETWORK LOAD	MEDIAN OBSERVED LATENCY	TELEMETRY ERROR	HISTORICAL JAM RATE	HARD SATURATION
18,000	**29.9%**	**153k ms**	**8.9%**	**8.2%**	**3**
12 links / 500 ticks	P95 61.3%	P95 465k ms	Median 3.0%	493 windows	257 latency nulls



What it explains: Shows where composite risk is concentrated. Thicker/redder links deserve earlier inspection.
Reasonable assumption: Use it for triage; trained-model outputs must justify any route override.



What it explains: Shows sustained load pressure. The average is 29.9%, below the 90% saturation threshold.
Reasonable assumption: An average can hide stressed links; pair it with the P95 ranking.



What it explains: Shows when disruptions cluster or decline; the overall jam rate is 8.22%.
Reasonable assumption: Spikes suggest changing threat intensity, not guaranteed live repetition.

Intelligence briefing

1 Latency becomes about 10.5x higher in the 80-90% band than in 10-20%.
2 Aegis-Elysium and Boreas-Fenix have trust-risk indices of 91% and 89%.
3 Top traffic-share deciles are jammed 2.6x as often as the bottom deciles.
4 Aegis-Boreas has the highest descriptive targeting risk and 11.8% jam rate.
5 Missing values: 257 observed latencies and 256 self-reports.

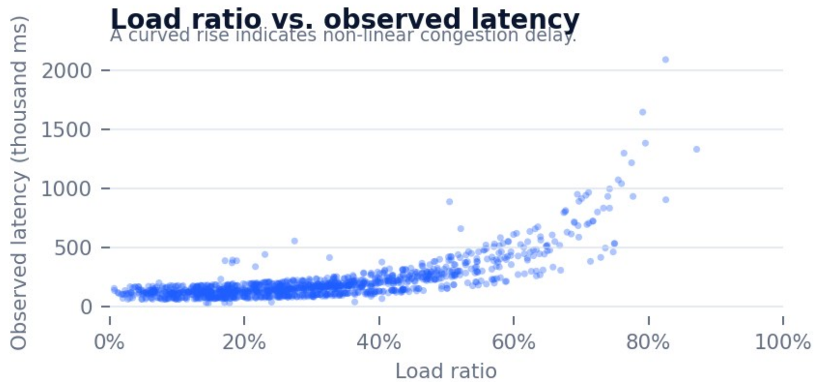
What it explains: Summarizes the strongest filter-driven facts and data-quality warnings.
Reasonable assumption: Every generated statement should trace back to a chart or metric.

VIEW 02

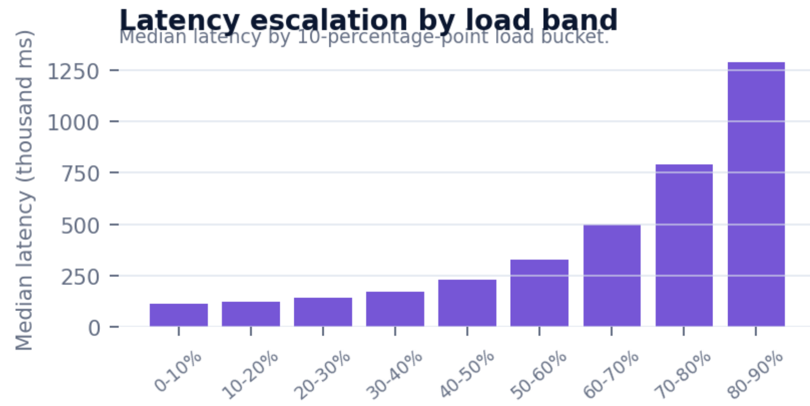
Congestion

Explains how link utilization converts into delay, stress, and eventual unavailability.

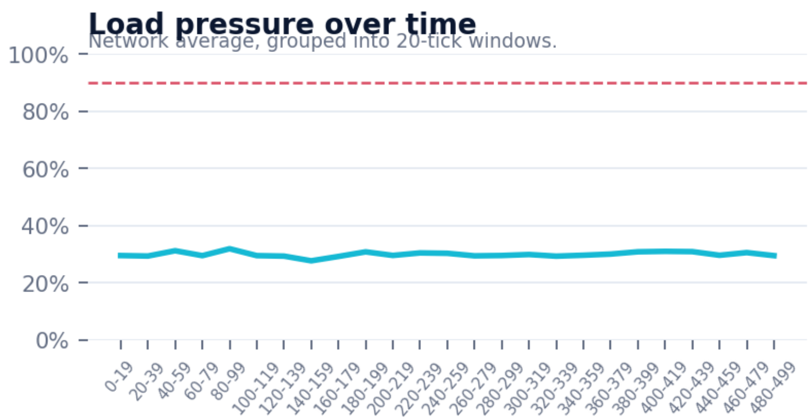
PEARSON RELATIONSHIP	SPEARMAN RELATIONSHIP	P95 OBSERVED LATENCY	UNAVAILABLE LATENCY
0.691	**0.713**	**465k ms**	**257**
Linear load-latency relationship	Monotonic relationship	Max 3.91M ms	Excluded from scatter and averages



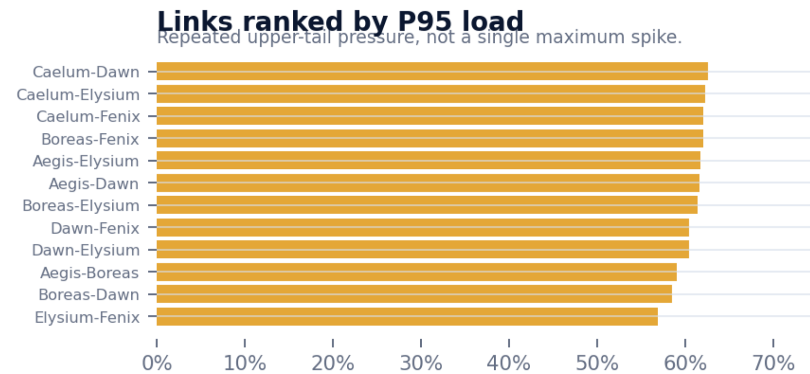
What it explains: Each point links current load ratio to measured latency. The upward curve and Pearson r=0.691 show a meaningful positive relationship.
Reasonable assumption: Load is a strong congestion feature, but other physical and environmental factors still explain part of latency.



What it explains: Median latency rises from 122k ms in 10-20% load to 1.29M ms in 80-90% load - about 10.5x.
Reasonable assumption: The congestion penalty should be non-linear; a simple constant penalty would understate high-load risk.



What it explains: Shows whether load pressure is temporary or sustained over time. No aggregate window reaches the 90% threshold.
Reasonable assumption: Network averages should not be used alone because saturation occurred in three individual link windows.



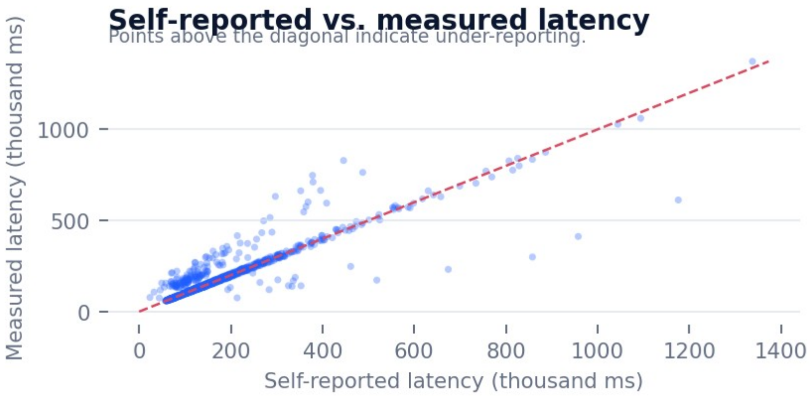
What it explains: Ranks links by P95 load, highlighting routes that repeatedly operate near their upper historical range. Caelum-Dawn has the highest P95 load at 62.6%.
Reasonable assumption: P95 is more reliable for repeated pressure than a single maximum; it should inform route cost and monitoring priority.

VIEW 03

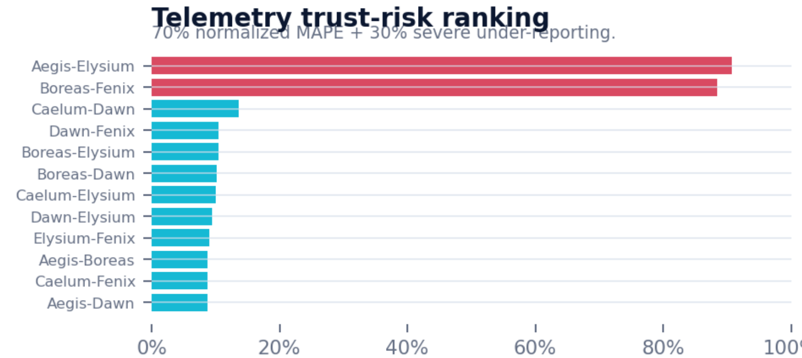
Telemetry trust

Explains whether a link reports latency honestly or consistently makes itself appear faster than reality.

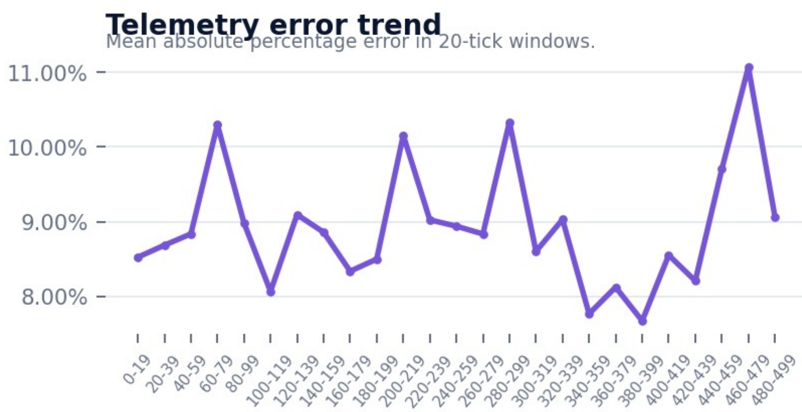
REPORTED-MEASURED CORR.	MEDIAN ABSOLUTE % ERROR	UNDER-REPORT FREQUENCY	MISSING SELF-REPORTS
0.934	**3.0%**	**55.9%**	**256**
Overall alignment	Mean 8.9%	Mean delta +10k ms	Excluded from trust calculations



What it explains: Compares what links claim with the ground truth. Points above the diagonal are slower than reported and therefore under-report actual latency.
Reasonable assumption: A high overall correlation can coexist with a few persistently deceptive links, so bias must be checked per link.



What it explains: Ranks links using normalized MAPE and severe under-reporting frequency. Aegis-Elysium and Boreas-Fenix are extreme outliers.
Reasonable assumption: Persistent, large directional error is more suspicious than ordinary symmetric reporting noise.



What it explains: Shows whether reporting error rises in particular historical windows. Full-history MAPE is 8.93%, while the median is only 2.99%.
Reasonable assumption: The gap between mean and median implies a small number of large discrepancies are pulling the average upward.

Links requiring investigation

#1 Aegis-Elysium — **9.3% trust**
MAPE 30.3% | severe under-reporting 72.2%

#2 Boreas-Fenix — **11.5% trust**
MAPE 29.3% | severe under-reporting 69.8%

#3 Caelum-Dawn — **86.4% trust**
MAPE 6.4% | severe under-reporting 1.9%

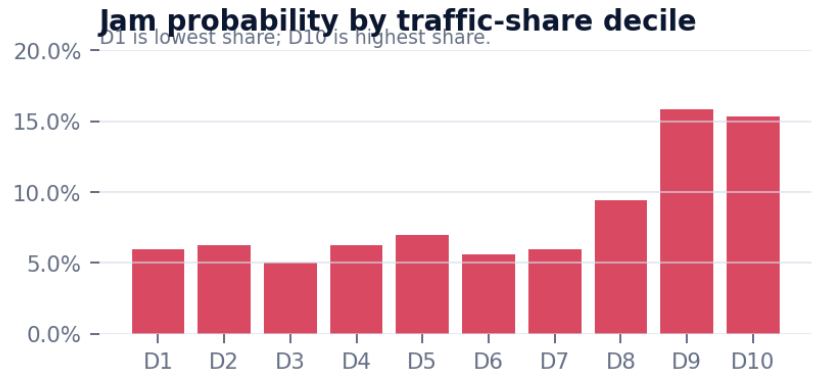
What it explains: Provides an audit-ready investigation list with MAPE, severe under-reporting, and resulting trust score.
Reasonable assumption: These links should receive a trust penalty or verification step, not be removed solely from one noisy observation.

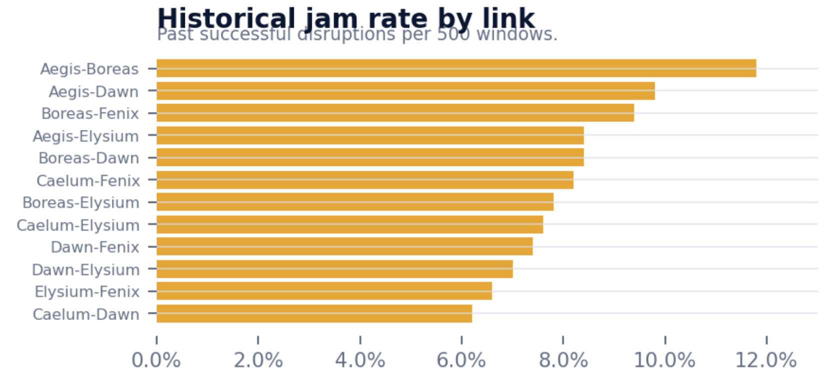
VIEW 04

Targeting risk

Explains whether Chimera preferentially disrupts heavily used and therefore predictable routes.

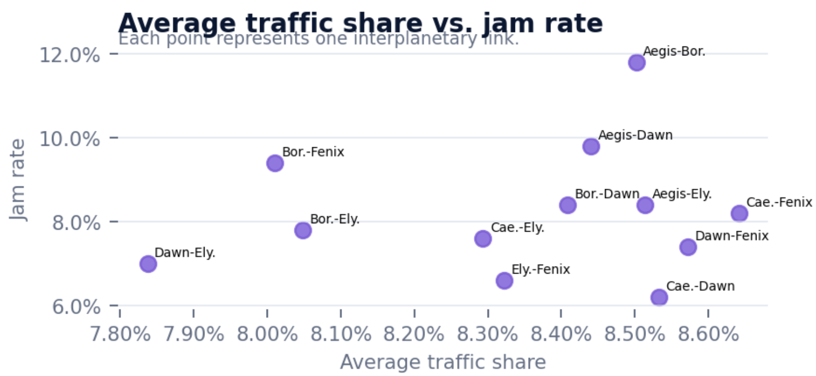
SUCCESSFUL DISRUPTIONS	SHARE WHEN JAMMED	SHARE-JAM RELATIONSHIP	VISIBLE LINK SET
493	11.4%	0.122	12
8.2% of windows	Safe windows 8.1%	Point-biserial correlation	Compare beyond one route

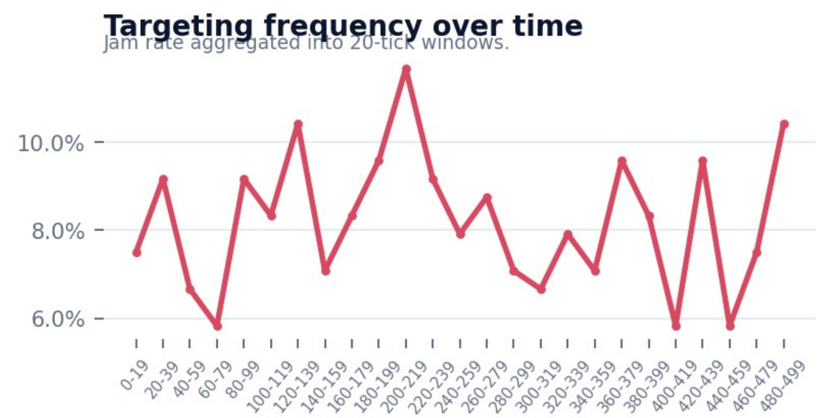




What it explains: Compares jam probability from the lowest to highest traffic-share windows. D9-D10 average 15.6% versus 6.1% for D1-D2 - about 2.6x higher.
Reasonable assumption: Traffic concentration is a meaningful targeting feature, but the relationship is probabilistic rather than deterministic.

What it explains: Shows which links were successfully disrupted most often. Aegis-Boreas leads with 59 jams and an 11.8% historical jam rate.
Reasonable assumption: Repeated route use can create a historical target signature; route diversification should reduce predictability.





What it explains: Compares each link average traffic share with its historical jam rate. The relationship is positive but modest (r=0.122).
Reasonable assumption: Traffic share should be combined with incident history and recent behavior, not used as the only targeting predictor.

What it explains: Shows whether jam activity changes across historical windows. Peaks indicate periods where the agent should increase caution or route entropy.
Reasonable assumption: A time trend can indicate regime changes, but the static CSV does not guarantee those regimes continue into live evaluation.

VIEW 05

Link intelligence

Combines congestion, trust, and targeting evidence into an auditable link-level profile.

HIGHEST COMPOSITE	NEXT HIGHEST	HIGHEST TARGETING	HIGHEST TRUST RISK
Boreas-Fenix	**Aegis-Elysium**	**Aegis-Boreas**	**Aegis-Elysium**
71.6% risk	71.2% risk	75.9%	90.7%

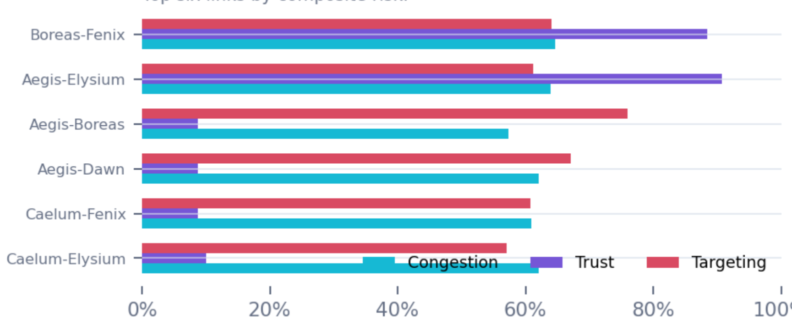
Unified link intelligence table

Link	Cong.	Trust	Target	Composite	Level
Boreas-Fenix	65%	89%	64%	72%	High
Aegis-Elysium	64%	91%	61%	71%	High
Aegis-Boreas	57%	9%	76%	48%	Moderate
Aegis-Dawn	62%	9%	67%	48%	Moderate
Caelum-Fenix	61%	9%	61%	45%	Moderate
Caelum-Elysium	62%	10%	57%	45%	Moderate
Caelum-Dawn	63%	14%	52%	45%	Moderate
Boreas-Elysium	60%	10%	57%	44%	Moderate
Boreas-Dawn	54%	10%	61%	43%	Moderate
Dawn-Fenix	57%	10%	57%	43%	Moderate
Dawn-Elysium	59%	9%	53%	43%	Moderate
Elysium-Fenix	53%	9%	53%	40%	Moderate

What it explains: The sortable table gives one comparable record per link: component risks, composite score, and risk level. The top two links are high risk; the remainder are moderate.
Reasonable assumption: Use sorting to locate the dominant threat type, not just the highest overall score.

Risk component breakdown

Top six links by composite risk.



What it explains: Separates the three threat components for the highest composite links. The top two are driven mainly by trust risk plus congestion evidence.
Reasonable assumption: Two links with similar composite scores may require different countermeasures if their component mixes differ.

Boreas-Fenix profile

Capacity	Average load	P95 latency
169 units	**30.4%**	**555k ms**
Trust score	Jam rate	Composite risk
11.5%	**9.4%**	**71.6%**

Interpretation: the high composite score is driven by both severe telemetry deception and repeated congestion/saturation evidence.

What it explains: The selected profile explains one route using capacity, load, latency, trust, jam history, and composite risk. Boreas-Fenix is the highest full-history composite case.
Reasonable assumption: A single profile supports a decision audit because each number is tied to a documented historical feature.

Explainable risk-index formula

01 Congestion P95 load + latency amplification + saturation + missing observations

02 Trust 70% normalized MAPE + 30% severe under-reporting

03 Targeting 65% jam rate + 35% average traffic share

04 Composite 40% congestion + 30% trust + 30% targeting

Use for dashboard triage only; the trained Phase 2 models should make live route decisions.

What it explains: Documents exactly how the descriptive risk index is calculated and weighted. It is intentionally simple and transparent.
Reasonable assumption: The index is for dashboard triage only; trained congestion, trust, and targeting models must produce the live route inputs.

Assumptions, limitations, and decision guidance

These conditions should be stated whenever the dashboard is presented or used to justify a routing decision.

VALID ASSUMPTIONS

- Load ratio is a useful predictor because latency rises strongly and non-linearly with load.
- Persistent directional telemetry error is a reasonable compromise signal.
- High traffic share increases targeting exposure at the population level.
- P95 metrics represent repeated stress better than a single maximum.
- Per-link component scores improve explainability and route comparison.

CAUTIONS

- Correlation does not establish causality.
- Null latency means unavailable or missing - never zero.
- Scatter charts sample points for readability, while metrics use all valid rows.
- Time charts aggregate 500 ticks into 25 windows, hiding within-window variation.
- Traffic-share deciles contain similar record counts, not equal share ranges.

LIVE-USE RULES

- Historical CSVs train and explain the models; they are not the live state.
- The build-period API values may be scrambled and should not retrain final models.
- A saturated link must be treated as unavailable.
- The final agent should combine physics cost with predicted congestion, trust, and targeting risk.
- Unseen behavior should trigger uncertainty handling rather than a confident guess.

DECISION-READY CONCLUSION

The five views support one consistent conclusion: congestion becomes sharply more expensive at high load, two links show strong historical telemetry deception, and concentrated traffic is targeted more often. The dashboard should therefore guide model inspection, route diversification, and audit explanations - while the trained Phase 2 models remain the source of live predictions.

Sources: link_traffic_history.csv, link_telemetry.csv, link_incident_history.csv, universe-config.json, DATASETS.md, and the LAUNCH26 Phase 02 challenge brief.